

$$\text{解 2: } G(s) = \frac{\Phi(s)}{1 - \Phi(s)} = \frac{\frac{K_0}{s(s+a)}}{1 - \frac{K_0}{s(s+a)}} = \frac{K_0}{s(s+a) - K_0} = \frac{K_0}{s^2 + as - K_0}$$

$$\begin{cases} K = 1 \\ v = 0 \end{cases} \quad \begin{aligned} K_p &= \lim_{s \rightarrow 0} G(s) = -1 \\ K_v &= \lim_{s \rightarrow 0} sG(s) = 0 \\ K_a &= \lim_{s \rightarrow 0} s^2 G(s) = 0 \end{aligned}$$

系统不稳定，求稳态误差无意义。

$$\text{解 3: } G(s) = \frac{K(\tau s + 1)}{s^v(Ts + 1)}$$

依题应有： $v > l$ (1)

$$D(s) = Ts^{v+1} + s^v + K\tau s + K = 0$$

$v \leq 2$ (2)

$l < v \leq 2$ (3)

当 $v=2$ 时： $D(s) = Ts^3 + s^2 + K\tau s + K = 0$

routh: ($v=2$)

s^3	T	$K\tau$	$\Rightarrow T > 0$
s^2	1	K	
s^1	$K(\tau - T)$		$\Rightarrow \tau > T$
s^0	K		$\Rightarrow K > 0$

$$T, K, \tau > 0, \quad l < v \leq 2$$

当 $v=1$ 时： $D(s) = Ts^2 + (1 + K\tau)s + K = 0$

$$T > 0, \quad K > 0, \quad \tau > -1/K, \quad l < v \leq 1$$